

# **Web Science**

## **Assignment – 9**

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# **1 Online advertisement**

In the videos about online advertisement, you learned about the three different payment methods for online advertisement. Consider the following meme spreading model.

1. Name all three methods. Which of the three payment models is the most similar to advertisement on TV?

**Answer:**

Three payment methods for an online advertisement:

1. Cost per mille.
2. Cost per acquisition
3. Cost per click

The cost per mille is a payment model is the most like advertisements on TV.

2. Mention the advantages and disadvantages, for publisher and advertiser, of all the payment methods and explain them in your own words.

**Answer:**

Advantages and Disadvantages of these payment methods:

1. CPM:

- a) The advantage of CPM for the publisher is that he doesn't have to care about the click rate on an ad if he has many visitors to his page.
- b) This is not so good for the advertiser as there is no certainty that the ad will be viewed or visited.

2. CPA:

- a) This is disadvantageous for the publisher as he gets paid only if the product is bought through the link on his page. So he might not make any money from some of his ads.
- b) This is advantageous for the advertiser as he has to pay the publisher only if the viewer has visited his store through the ad and has bought the product.

3. CPC:

- a) The advantage for the publisher is that he doesn't have to worry if the user buys the product.

b) The advantage for the advertiser is that he pays only for the clicks and once the user clicks on the ad, the better the product, the more the chances of it being sold.

3. Provide real world examples for the three payment methods.

**Answer:**

1. Cost per mille: The ads in the newspaper, television, and radio are the best examples of this. Every thousand views are paid by an agreed amount of money to the publisher.

2. Cost per acquisition: Newsletter sign-ups, contact requests, and online registration are the best examples of these. Money paid to the publisher gets paid only when there is a follow-up action through the ad. : when users buy something after clicking on the ad.

3. Cost per click: Adds on Google are the best examples of these. Money is paid for the number of clicks, or the click rate on the ad.

## **2 Payments in Online Advertisement**

Provide the complete calculation with your solutions for the following questions.

1. An online advertisement company offers you to advertise your website on a cost-per-click base (CPC) with a cost of €1.20 per click. Assuming that in average two out of ten visitors of your website buy a product from which you are earning 20€, would you accept this offer? What is your average profit/loss per visitor?

**Answer:**

Let us assume 10 visitor visits our website and 2 of them buy the product.

Cost per click = €1.20

Total Visitors = 10

Product brought visitor = 2

Total spent =  $10 * 1.2 = €12$

Total earned =  $2 * 20€ = €40$

Total spent < Total earned

Total profit =  $€40 - €12 = €28$

Average profit per visitor =  $€28/10 = €2.8$

Hence, we would accept the offer.

2. What would be the minimal conversion rate (CR) to guarantee your profit?

**Answer:**

We know that,

Customer Per Click (CPC) = Customer Acquisition Costs (CAC) \* Conversion Rate (CR)

$CPC = CAC * CR$

$1.2 = 20 * CR$

$CR = 1.2/20$

$CR = 0.06$

The above scenario is considered when there is one visitor and one visitor buying the product. So, in order to guarantee profit the minimum conversion rate should not be less than 0.06 or 6%.

3. Two online advertisement companies A and B are making you offers to advertise your website. Company A follows a cost-per-mille (CPM) model with a cost of 2.40€ for displaying your banner advertisement thousand times. Company B follows a cost-per-action (CPA) model charging a commission of 6% from every profit generated on your website through clicks on the banner ad. Assuming a click-through-rate (CTR) of 0.5%, a conversion rate (CR) of 20% and an average profit of 40€ for every transaction on your website, which offer is the best?

**Answer:**

**Company A (CPM)**

Cost = 2.40 € for 1000 views

Click-through-rate (CTR) = 0.5%

Conversion rate (CR) = 20%

Average profit = 40€ for every transaction

We know that,

$CTR = \text{Number of clicks} / \text{Number of views}$

$0.5/100 = \text{Number of clicks} / 1000$

$\text{Number of clicks} = 0.5/100 * 1000 = 5$

$CR = \text{Number of sales} / \text{Number of clicks}$

$20/100 = \text{Number of sales} / 5$

$\text{Number of sales} = 20/100 * 5 = 1$

Each sale profit =  $1 * 40€ = 40€$

Total overall profit =  $40€ - 2.40€ = 37.6€$

### **Company B (CPA)**

Commission rate = 6%

Each sale = 40 €

Commission given =  $6/100 * 40 = 2.4\text{€}$

Total overall profit =  $40\text{€} - 2.40\text{€} = 37.6\text{€}$

Hence, from both companies, the overall profit generated is the same. Keeping the given scenario, we can choose any company. But the best offer is CPA where if there are not many sales then we don't have to pay more commission.

4. Assuming an online advertisement campaign for a website has obviously a high click-through rate (CTR), but the earnings from the website are still very poor. What do you think could be the problem (please provide your answer in one or two paragraphs)?

**Answer:** If an advertisement has a high click-through rate, however, the earnings are low implies that the advertisement is placed for the targeted customers and the ads are attractive which is gaining popularity for high clicks. Despite high click, when the earnings is low refer to the low conversion rate. 

There are different reasons for the low conversion rate:

- The visitor to the website is not the actual/real customer.
- The website has some problems with the design or other issues like lagging or difficulty in understanding which makes the user leave the website sooner.
- The advisement is maybe kind of fraud advertisement.
- The advisement and the result after landing on the website might differ.